

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Standard & Compound Units

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STANDARD & COMPOUND UNITS · CH4 · L1



ENG. ESLAM AHMED

The Map

This lesson collapses Standard & Compound Units into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q13

Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The cube rests on one

$$w^2 = 400$$

Finish: 8000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A cube rests on one face. The force on the table is 72 N and the pressure is 0.18 N/cm^2 . Find the volume of the cube.

YOUR SOLUTION

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{mass} = \text{density} \times \text{volume}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q9

Calculate volume

$$\text{mass} = \text{density} \times \text{volume}$$

Finish: 2895 g.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q5

The time is

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

Average speed

$$2470 \div 3.6 = 686.111 \dots$$

Finish: 686 km/h.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

| A cyclist travels 18 km in 45 minutes. Work out the average speed in km/h.

YOUR SOLUTION

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1 \text{ m} = 100 \text{ cm}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q2

Calculate value

$$1 \text{ m} = 100 \text{ cm}$$

Calculate value

$$1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

Finish: 32,400,000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Pressure: force over area
"work out", "show", "hence", a familiar exam phrase	Density: mass and volume
"work out", "show", "hence", a familiar exam phrase	Speed: distance over time
"work out", "show", "hence", a familiar exam phrase	Volume unit conversion

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Angles in Polygons & Parallel Lines

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ANGLES IN POLYGONS & PARALLEL LINES · CH4 · L2

ENG. ESLAM AHMED



The Map

This lesson collapses Angles in Polygons & Parallel Lines into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 4 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

4 The diagram shows a triangle. Diagram NOT accurately drawn 30 degrees ($4x + 10$) degrees ($x + 20$) degrees Work out the value of x . $x =$

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(9x + 27) + (2x - 23) = 180$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 P1H Q12

The interior angle and exterior

$$(9x + 27) + (2x - 23) = 180$$

The exterior angle is

$$2x - 23 = 2(16) - 23 = 9^\circ$$

Finish: $n = 40$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

177, 175, 173, 171, 169, ...

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

177, 175, 173, 171, 169, ...

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

23 A polygon has n sides, where $n > 5$. When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence. Here are the first five terms of this sequence: 177, 175, 173, 171, 169. Find the value of n . Show clear algebraic working. Question 23 continued n =

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{(5-2)180}{5} = 108^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q8

A regular pentagon has interior

$$\frac{(5-2)180}{5} = 108^\circ$$

A regular hexagon has interior

$$\frac{(6-2)180}{6} = 120^\circ$$

Finish: 96° .

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 The diagram shows a regular pentagon, ABCDE, a regular hexagon, CFGHID, and a quadrilateral, EDIJ. A B F G H C D I x J E Diagram NOT accurately drawn AEJ and HIJ are straight lines. Work out the size of the angle marked x. Show your working clearly. degrees

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

177, 175, 173, 171, 169, ...

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

177, 175, 173, 171, 169, ...

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: 360° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "find", a missing value	Find the gradient
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Bearings, Scale Drawing & Constructions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

BEARINGS, SCALE DRAWING & CONSTRUCTIONS · CH4 · L3

ENG. ESLAM AHMED



The Map

This lesson collapses Bearings, Scale Drawing & Constructions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1 cm represents 200 m

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q5

The distance from to is

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale is

1 cm represents 200 m

Finish: Draw C 3.5 cm from A on a bearing of 110° , and the scale is 1 : 20000.

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 The scale diagram shows the position on a map of a house, A A North Scale: 1 cm represents 200 metres House C is on a bearing of 110 degrees from A The distance from A to C is 700 m (a) Mark the position of C on the diagram with a cross (times) Label your cross C (3) (b) Write the scale of the map in the form 1 : n 1 : (1)

YOUR SOLUTION

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw North first
"bearing", "north", "scale drawing"	Measure clockwise bearing
"bearing", "north", "scale drawing"	Use scale before real distance
"draw", "plot", "construct", a dia- gram/grid	Construct the locus
"bearing", "north", "scale drawing"	Use bearings
"draw", "plot", "construct", a dia- gram/grid	Construction answer

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circle Theorems

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLE THEOREMS · CH4 · L4

ENG. ESLAM AHMED



The Map

This lesson collapses Circle Theorems into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 5 local PDFs read.

01 Spot the theorem family

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

02 Use tangent-radius or alternate segment

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle BFD = 39^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q14

Use trigonometry

$$\angle BFD = 39^\circ$$

Given

$$\angle EFD = 18^\circ$$

Finish: 123° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use cyclic quadrilateral angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P2H Q18

Use trigonometry

$$AC^2 = 8^2 + 7.5^2 - 2(8)(7.5) \cos 98^\circ$$

Opposite angles in a cyclic

$$\angle ADC = 180^\circ - 98^\circ = 82^\circ$$

Finish: 32.8 cm.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

04 Write the reason with the answer

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle RQP = 90^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q14

Use trigonometry

$$\angle RQP = 90^\circ$$

Reason the angle in a

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\angle RQP = 90^\circ$, and $\angle PSQ = 22^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q14

Angle and central angle stand

This is a reasoning step: write one clear sentence, then continue the calculation.

The angle at the centre

$$\angle AOC = 2 \times 38^\circ = 76^\circ$$

Finish: 52° .

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

06 Use circle theorem

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circle Theorems question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m = \frac{124}{2} = 62$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q13

Use trigonometry

$$m = \frac{124}{2} = 62$$

Use circle theorem

$$n = 180 - 62 = 118$$

Finish: $m = 62$, $n = 118$, and $\angle QPO = 62^\circ$.

FORMULA BANK

Angle at centre = twice angle at circumference. Tangent-radius angle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

In a circle, A, B, C, D lie on the circumference and AB is a tangent at A . One given angle is 58° . Find the missing angle and give a reason.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Spot the theorem family
"work out", "show", "hence", a familiar exam phrase	Use tangent-radius or alternate segment
"work out", "show", "hence", a familiar exam phrase	Use cyclic quadrilateral angles
"work out", "show", "hence", a familiar exam phrase	Write the reason with the answer
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use circle theorem
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Perimeter

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & PERIMETER · CH4 · L5

ENG. ESLAM AHMED



The Map

This lesson collapses Area & Perimeter into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Split the compound shape

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Build an equation from area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use perimeter after missing sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check units squared or linear

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 32 website questions, 32 checked solutions, 4 local PDFs read.

01 Split the compound shape

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two

This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Build an equation from area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = \frac{1}{2}\pi r^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q5

The garden is a semicircle

$$\text{area} = \frac{1}{2}\pi r^2$$

Calculate area

$$12 \times 6 = 72 \text{ m}^2$$

Finish: No, he does not have enough grass seed.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 The diagram shows Yuen's garden. 7.2 m Diagram NOT accurately drawn The garden is in the shape of a semicircle of radius 7.2 m. Yuen is going to cover his garden with grass seed. Yuen has 12 boxes of grass seed. Each box of grass seed contains enough seed to cover 6 m² of the garden. Has Yuen enough grass seed for his garden? Show your working clearly.

YOUR SOLUTION

03 Use perimeter after missing sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$12 \times 9 = 108 \text{ cm}^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q3

Split the shape into two

This is a reasoning step: write one clear sentence, then continue the calculation.

The left rectangle has area

$$12 \times 9 = 108 \text{ cm}^2$$

Finish: 7 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

3 The diagram shows a shape. 9 cm x cm 6 cm 12 cm Diagram NOT accurately drawn
The shape has area 129 cm² Work out the value of x. x =

YOUR SOLUTION

04 Check units squared or linear

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AD = 17 \text{ cm}, \quad CD = 15 \text{ cm}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q11

Use trigonometry

$$AD = 17 \text{ cm}, \quad CD = 15 \text{ cm}$$

Use Pythagoras to find

$$AC^2 = 17^2 - 15^2 = 289 - 225 = 64$$

Finish: 44.6 cm.

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

11 The diagram shows a shaded shape ABCD made from a semicircle ABC and a right-angled triangle ACD. Diagram NOT accurately drawn

17 cm 15 cm

A B C D

AC is the diameter of the semicircle ABC. Work out the perimeter of the shaded shape. Give your answer correct to 3 significant figures. cm

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Perimeter question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$PQ = p$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q20

Let

$$PQ = p$$

Given

$$AB = 1.5PQ$$

Finish: $PQ = 4.8 \text{ cm}$

FORMULA BANK

Break the shape. Rectangle lw , triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

20 A B F P Q C D R E Diagram NOT accurately drawn T The diagram shows a shaded region T formed by removing an equilateral triangle PQR from a regular hexagon ABCDEF. The points P and Q lie on AB such that $AB = 1.5$ times PQ Given that the area of region T is 72.3 cm^2 work out the length of PQ. cm

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Split the compound shape
"work out", "show", "hence", a familiar exam phrase	Build an equation from area
"work out", "show", "hence", a familiar exam phrase	Use perimeter after missing sides
"work out", "show", "hence", a familiar exam phrase	Check units squared or linear
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circles, Arcs & Sectors

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLES, ARCS & SECTORS · CH4 · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Circles, Arcs & Sectors into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find sector area

TRIGGER *"work out", "find", a missing value*

03 Find arc length

TRIGGER *"work out", "find", a missing value*

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 41 website questions, 41 checked solutions, 4 local PDFs read.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q8

The inner radius is m

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$7 + 2 = 9 \text{ m}$$

Finish: 54.3 m.

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

02 Find sector area

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q8

The inner radius is m

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$7 + 2 = 9 \text{ m}$$

Finish: 54.3 m.

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
 2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

03 Find arc length

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5 = \frac{55}{360} \times 2\pi r$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q16

Use trigonometry

$$5 = \frac{55}{360} \times 2\pi r$$

Area of the circle

$$\pi r^2 = \pi(5.208\dots)^2 = 85.233\dots$$

Finish: 85.2 cm².

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x. Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x . Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use fraction of the circle
"work out", "find", a missing value	Find sector area
"work out", "find", a missing value	Find arc length
"work out", "show", "hence", a familiar exam phrase	Combine sector with triangle/rectangle
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Volume & Surface Area

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

VOLUME & SURFACE AREA · CH4 · L7

ENG. ESLAM AHMED



The Map

This lesson collapses Volume & Surface Area into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Name the solid

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

FORMULA BANK

Prism volume = area of cross-section × length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

02 Use prism volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{2}(7 + 12)h = 9.5h$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q13

Area of the trapezium cross

$$\frac{1}{2}(7 + 12)h = 9.5h$$

Volume of the prism

$$9.5h \times 10 = 608$$

Finish: $h = 6.4$ cm.

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

13 Diagram NOT accurately drawn A B D C 7 cm E 10 cm 12 cm h cm F G H The diagram shows a prism ABCDEFGH in which ABCD is a trapezium with BC parallel to AD and CDEF is a rectangle. BC = 7 cm AD = 12 cm DE = 10 cm The height of trapezium ABCD is h cm The volume of the prism is 608 cm³ Work out the value of h. h =

YOUR SOLUTION

03 Unfold surface area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

04 Reverse from volume or area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{3}\pi r^2 h = 1600$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q26

Let the radius of the

This is a reasoning step: write one clear sentence, then continue the calculation.

The volume of the cone

$$\frac{1}{3}\pi r^2 h = 1600$$

Finish: 107°

FORMULA BANK

Prism volume = area of cross-section $\times$ length. Surface area = sum of visible faces.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

26 Here is a sector, AOB, of a circle with centre O and angle AOB = x degrees. Diagram NOT accurately drawn. The sector can form the curved surface of a cone by joining OA to OB. Diagram NOT accurately drawn. The height of the cone is 25 cm. The volume of the cone is 1600 cm³. Work out the value of x . Give your answer correct to the nearest whole number. ' x =

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Volume & Surface Area question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q17

The cross section is a

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Calculate area

$$\frac{37 - 28}{2} = 4.5$$

Finish: 3844 cm², and 39.3°.

FORMULA BANK

Prism volume = area of cross-section × length. Surface area = sum of visible faces.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

17 The diagram shows a solid prism ABCDEFGH. A 28 cm 24 cm 20 cm D E F G H B C Diagram NOT accurately drawn 37 cm The trapezium ABCD, in which AD is parallel to BC, is a cross section of the prism. The base ADEH of the prism is a horizontal plane. ADEH and BCFG are rectangles. The midpoint of BC is vertically above the midpoint of AD so that BA = CD. AD = 37 cm GF = 28 cm DE = 24 cm The perpendicular distance between edges AD and BC is 20 cm. (a) Work out the total surface area of the prism. cm² (4) (b) Calculate ...

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the solid
"work out", "show", "hence", a familiar exam phrase	Use prism volume
"work out", "show", "hence", a familiar exam phrase	Unfold surface area
"work out", "show", "hence", a familiar exam phrase	Reverse from volume or area
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Congruence, Similarity & Geometrical Proof

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CONGRUENCE, SIMILARITY & GEOMETRICAL PROOF · CH4 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Congruence, Similarity & Geometrical Proof into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 11 website questions, 11 checked solutions, 4 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 Diagram NOT accurately drawn A (2x + 9) cm 2x cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$CD = 1.5BE$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$CD = 1.5BE$$

Finish: (a) $ED = 5$ cm, (b) $x = 3.75$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 10 cm Diagram NOT accurately drawn E D C A B In the diagram, ABC and AED are straight lines. BE is parallel to CD AE = 10 cm and CD = 1.5 times BE (a) Work out the length of ED cm (2) AB = (2x + 5) cm and BC = (3x - 5) cm (b) Work out the value of x x = (2)

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q3

The scale factor from triangle

$$\frac{PQ}{AB} = \frac{12}{4} = 3$$

Evaluate fraction

$$BC = \frac{RQ}{3} = \frac{16.5}{3} = 5.5$$

Finish: $BC = 5.5$ cm, and $y = 3x$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

3 Diagram NOT accurately drawn A R Q P C B 4 cm x cm 12 cm y cm 16.5 cm Triangle ABC is similar to triangle PQR AB = 4 cm PQ = 12 cm RQ = 16.5 cm AC = x cm PR = y cm (a) Calculate the length of BC cm (2) (b) Write down an expression for y in terms of x y = (1)

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Congruence, Similarity & Geometrical Proof question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{AC}{CE} = \frac{BC}{CD}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q12

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$\frac{AC}{CE} = \frac{BC}{CD}$$

Finish: $x = \frac{9}{8}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 Diagram NOT accurately drawn A (2x + 9) cm 2x cm 1.2 cm 6 cm D E B C ACE and BCD are straight lines. AB is parallel to DE Work out the value of x x =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Evaluate fraction
"work out", "find", a missing value	Find the gradient
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Area & Volume of Similar Shapes

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

AREA & VOLUME OF SIMILAR SHAPES · CH4 · L9

ENG. ESLAM AHMED



The Map

This lesson collapses Area & Volume of Similar Shapes into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BANK EVIDENCE 36 website questions, 36 checked solutions, 3 local PDFs read.

01 Find the length scale factor

TRIGGER *"work out", "find", a missing value*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

$$15 : 45 = 1 : 3$$

Finish: 5090 cm²

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn

15 cm 39 cm 39 cm 45 cm

The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm²

YOUR SOLUTION

02 Square it for area

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$15 : 45 = 1 : 3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2023 P1H Q23

The small and large square

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are

$$15 : 45 = 1 : 3$$

Finish: 5090 cm²

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram. Diagram NOT accurately drawn

15 cm 39 cm 39 cm 45 cm

The height of the small pyramid is 15 cm. The height of the large pyramid is 45 cm. The square base of the large pyramid has side length 39 cm. Work out the total surface area of the frustum. Give your answer correct to the nearest whole number. cm²

YOUR SOLUTION

03 Cube it for volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q20

The two cones are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

The large cone has height

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

20 The diagram shows a frustum of a cone and a sphere. The frustum is made by removing a small cone from a large cone. The cones are similar. h cm $2h$ cm r cm r cm
Diagram NOT accurately drawn The height of the small cone is h cm. The height of the large cone is $2h$ cm. The radius of the base of the large cone is r cm. The radius of the sphere is r cm. Given that the volume of the frustum is equal to the volume of the sphere, find an expression for r in terms of h . Give your expression in its simplest form. $r =$

YOUR SOLUTION

04 Reverse from area or volume

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{h}{2h} = \frac{1}{2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q20

The two cones are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

The large cone has height

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $r = \frac{7h}{16}$

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

20 The diagram shows a frustum of a cone and a sphere. The frustum is made by removing a small cone from a large cone. The cones are similar. h cm $2h$ cm r cm r cm
Diagram NOT accurately drawn The height of the small cone is h cm. The height of the large cone is $2h$ cm. The radius of the base of the large cone is r cm. The radius of the sphere is r cm. Given that the volume of the frustum is equal to the volume of the sphere, find an expression for r in terms of h . Give your expression in its simplest form. $r =$

YOUR SOLUTION

05 Split the ratio

TRIGGER *"similar", "scale factor", "area/volume"*

BECOMES A Area & Volume of Similar Shapes question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $9 : 13$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q20

The two vases are similar

This is a reasoning step: write one clear sentence, then continue the calculation.

Their heights are in the

$9 : 13$

Finish: 583.2 cm^2

FORMULA BANK

Length scale factor k . Area scale factor k^2 . Volume scale factor k^3 .

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

20 The diagram shows two similar vases, A and B. Diagram NOT accurately drawn
 9 cm 13 cm A B The height of vase A is 9 cm and the height of vase B is 13 cm. Given
 that surface area of vase A + surface area of vase B = 1800 cm² calculate the surface
 area of vase A. cm²

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find the length scale factor
"similar", "scale factor", "area/volume"	Square it for area
"similar", "scale factor", "area/volume"	Cube it for volume
"similar", "scale factor", "area/volume"	Reverse from area or volume
"similar", "scale factor", "area/volume"	Split the ratio

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Right-Angled Triangles - Pythagoras & Trigonometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RIGHT-ANGLED TRIANGLES - PYTHAGORAS & TRIGONOMETRY · CH4 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Right-Angled Triangles - Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$EF = 12 \cos 40^\circ = 9.192 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q15

Use trigonometry

$$EF = 12 \cos 40^\circ = 9.192 \dots$$

Use trigonometry

$$\sin 28^\circ = \frac{EF}{EG}$$

Finish: 19.6 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BD = 16, \quad \angle DAB = 65^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q10

Find the gradient

This is a reasoning step: write one clear sentence, then continue the calculation.

Use trigonometry

$$BD = 16, \quad \angle DAB = 65^\circ$$

Finish: 50.2 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\cos 63^\circ = \frac{24.3}{PQ}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q9

Use trigonometry

$$\cos 63^\circ = \frac{24.3}{PQ}$$

Finish: 53.5 cm.

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

06 Angle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Right-Angled Triangles - Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB^2 = 21^2 - 15^2 = 216$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q11

Use trigonometry

$$AB^2 = 21^2 - 15^2 = 216$$

Angle

$$\cos ACB = \frac{15}{21}$$

Finish: 13.5° .

FORMULA BANK

SOH CAH TOA and $a^2 + b^2 = c^2$. Choose from the labelled sides.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

A right-angled triangle has hypotenuse 14.2 cm and angle 37° . Find the side opposite the angle.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Angle

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sine, Cosine Rule & Area of Triangles

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SINE, COSINE RULE & AREA OF TRIANGLES · CH4 · L11

ENG. ESLAM AHMED



The Map

This lesson collapses Sine, Cosine Rule & Area of Triangles into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$BC = 13 \sin 40^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q21

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

Find the gradient

$$BC = 13 \sin 40^\circ$$

Finish: 12.4 cm

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q14

(a)

$$\text{area} = 9.3 \times 14.7 \times \sin 106^\circ = 131.414 \dots$$

Answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 19.4 cm.

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{area} = \frac{1}{2}ab \sin C.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

06 Use cosine rule

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Sine, Cosine Rule & Area of Triangles question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{BC}{2} = 14$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q9

Use cosine rule

$$\frac{BC}{2} = 14$$

Finish: 147 cm².

FORMULA BANK

$$\frac{a}{\sin A} = \frac{b}{\sin B}, a^2 = b^2 + c^2 - 2bc \cos A, \text{ area} = \frac{1}{2}ab \sin C.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

| In triangle ABC , $AB = 9.4$ cm, $AC = 7.1$ cm and angle $A = 48^\circ$. Find BC .

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "show", "hence", a familiar exam phrase	Use cosine rule

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

3D Pythagoras & Trigonometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

3D PYTHAGORAS & TRIGONOMETRY · CH4 · L12

ENG. ESLAM AHMED



The Map

This lesson collapses 3D Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

03 Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the midpoint

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q22

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The line from to the

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

Finish: 74.7°

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q22

Use trigonometry

This is a reasoning step: write one clear sentence, then continue the calculation.

The line from to the

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

Finish: 74.7°

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q19

The projection of onto the

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

Use trigonometry

$$\tan \theta = \frac{6}{\sqrt{106}}$$

Finish: 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

04 Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB = 15, \quad BC = 4, \quad CF = 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q22

Find the gradient

$$AB = 15, \quad BC = 4, \quad CF = 9$$

Use trigonometry

$$PR = 42$$

Finish: $BE = 17.9 \text{ cm}$, $\angle UMR = 49.1^\circ$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q19

The projection of onto the

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

Use trigonometry

$$\tan \theta = \frac{6}{\sqrt{106}}$$

Finish: 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Find the midpoint

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AE = x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q21

Let

$$AE = x$$

Use trigonometry

$$AB = BD = DE = AE = x$$

Finish: $T = \frac{\sqrt{68} + \sqrt{51}}{17}$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw the hidden right triangle
"work out", "find", a missing value	Find the base diagonal first
"work out", "show", "hence", a familiar exam phrase	Use 3D trig after Pythagoras
"work out", "show", "hence", a familiar exam phrase	Give the final angle/length
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
"work out", "find", a missing value	Find the midpoint

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.